

ECOSYSTEM STRUCTURE AND FUNCTION

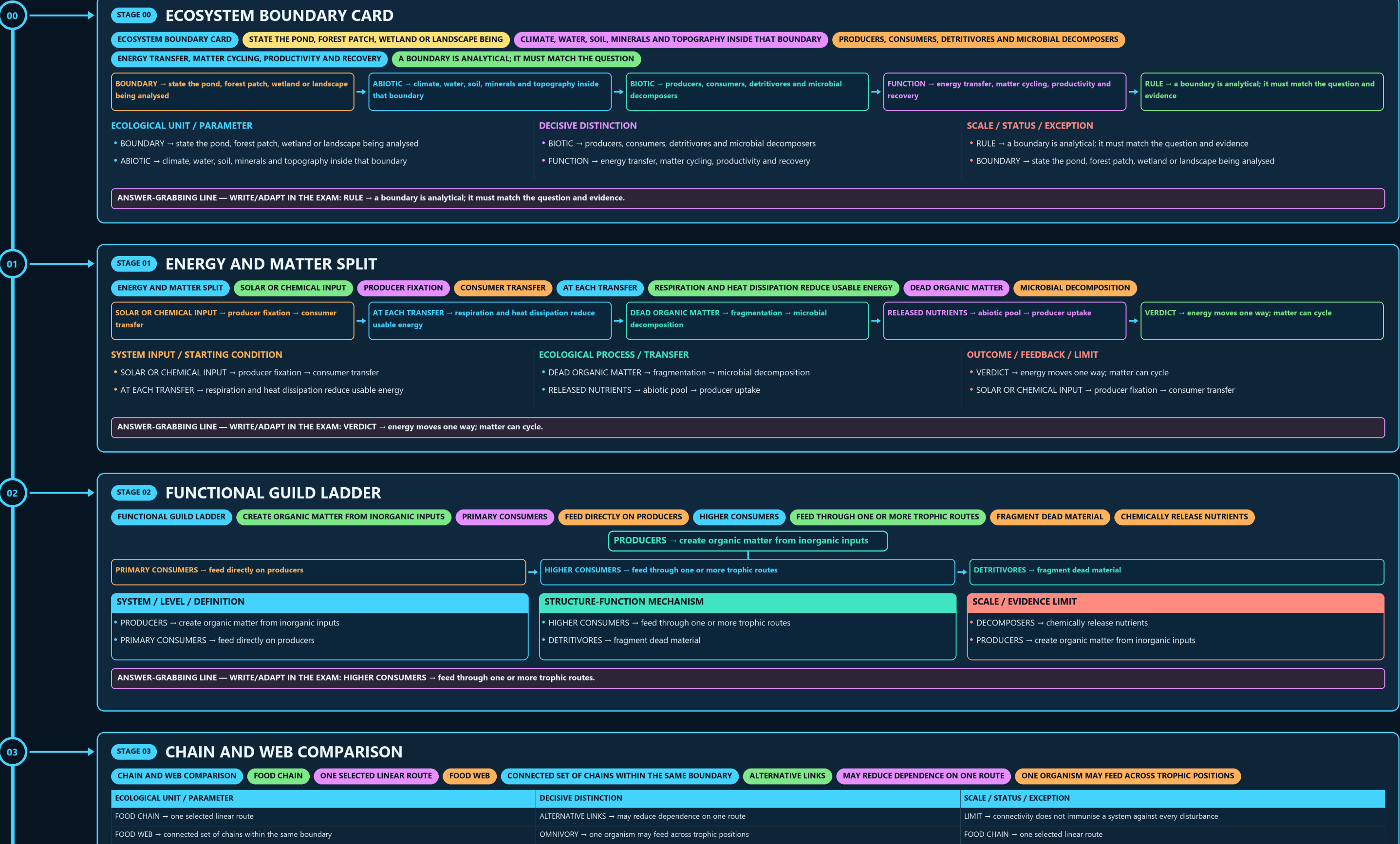
ECOSYSTEM BOUNDARY CARD → ENERGY AND MATTER SPLIT → FUNCTIONAL GUILD LADDER → CHAIN AND WEB COMPARISON → HABITAT AND NICHE MATRIX → ECOTONE LOGIC

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

• PRIMARY CONSUMERS → feed directly on producers

• DETRITIVORES → fragment dead material

• PRODUCERS → create organic matter from inorganic inputs

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIGHER CONSUMERS → feed through one or more trophic routes.

STAGE 03CHAIN AND WEB COMPARISON

CHAIN AND WEB COMPARISONFOOD CHAINONE SELECTED LINEAR ROUTEFOOD WEBCONNECTED SET OF CHAINS WITHIN THE SAME BOUNDARYALTERNATIVE LINKSMAY REDUCE DEPENDENCE ON ONE ROUTEONE ORGANISM MAY FEED ACROSS TROPHIC POSITIONS

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
FOOD CHAIN → one selected linear route	ALTERNATIVE LINKS → may reduce dependence on one route	LIMIT → connectivity does not immunise a system against every disturbance
FOOD WEB → connected set of chains within the same boundary	OMNIVORY → one organism may feed across trophic positions	FOOD CHAIN → one selected linear route

ECOLOGICAL UNIT / PARAMETER

• FOOD CHAIN → one selected linear route

• FOOD WEB → connected set of chains within the same boundary

DECISIVE DISTINCTION

• ALTERNATIVE LINKS → may reduce dependence on one route

• OMNIVORY → one organism may feed across trophic positions

SCALE / STATUS / EXCEPTION

• LIMIT → connectivity does not immunise a system against every disturbance

• FOOD CHAIN → one selected linear route

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIMIT → connectivity does not immunise a system against every disturbance.

04

HABITAT AND NICHE MATRIX

PHYSICAL ADDRESSHABITAT NICHEWHERE WITHIN THAT ADDRESS THE ORGANISM OPERATESTROPHIC NICHERESOURCES AND FEEDING RELATIONSREPRODUCTIVE NICHEBREEDING REQUIREMENTS AND TIMING

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
HABITAT → physical address	TROPHIC NICHE → resources and feeding relations	RULE → overlap in place does not prove identical functional niches
HABITAT NICHE → where within that address the organism operates	REPRODUCTIVE NICHE → breeding requirements and timing	HABITAT → physical address

ECOLOGICAL UNIT / PARAMETER

• HABITAT → physical address

• HABITAT NICHE → where within that address the organism operates

DECISIVE DISTINCTION

• TROPHIC NICHE → resources and feeding relations

• REPRODUCTIVE NICHE → breeding requirements and timing

SCALE / STATUS / EXCEPTION

• RULE → overlap in place does not prove identical functional niches

• HABITAT → physical address

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → overlap in place does not prove identical functional niches.

05

STAGE 05ECOTONE LOGIC

ECOTONE LOGICECOSYSTEM ATRANSITION ZONE OF VARIABLE WIDTHECOSYSTEM BOVERLAPPING CONDITIONSPECIES FROM BOTH SIDES MAY OCCURNEW EDGE CONDITIONSSOME ADDITIONAL SPECIALISTS MAY OCCUR

ECOSYSTEM A → transition zone of variable width → ECOSYSTEM B

OVERLAPPING CONDITIONS → species from both sides may occur

NEW EDGE CONDITIONS → some additional specialists may occur

COMPETITION AND TENSION → the zone is not automatically benign

EDGE EFFECT → possible density or variety pattern, not the definition

SYSTEM / LEVEL / DEFINITION	STRUCTURE-FUNCTION MECHANISM	SCALE / EVIDENCE LIMIT
• ECOSYSTEM A → transition zone of variable width → ECOSYSTEM B • OVERLAPPING CONDITIONS → species from both sides may occur	• NEW EDGE CONDITIONS → some additional specialists may occur • COMPETITION AND TENSION → the zone is not automatically benign	• EDGE EFFECT → possible density or variety pattern, not the definition • ECOSYSTEM A → transition zone of variable width → ECOSYSTEM B

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EDGE EFFECT → possible density or variety pattern, not the definition.

06

STAGE 06FOREST STRATIFICATION SECTION

FOREST STRATIFICATION SECTIONHIGH LIGHT, WIND EXPOSURE AND ARBOREAL NICHESFILTERED LIGHT AND HUMID MICROCLIMATESHRUB AND HERB LAYERSNEAR-GROUND RESOURCE PARTITIONINGFOREST FLOORLITTER, DETRITIVORES AND DECOMPOSER ACTIVITY

MORE LAYERS DO NOT PROVE UNIVERSAL STABILITY

CANOPY → high light, wind exposure and arboreal niches

UNDERSTOREY → filtered light and humid microclimate

SHRUB AND HERB LAYERS → near-ground resource partitioning

FOREST FLOOR → litter, detritivores and decomposer activity

SYSTEM / LEVEL / DEFINITION	STRUCTURE-FUNCTION MECHANISM	SCALE / EVIDENCE LIMIT
• CANOPY → high light, wind exposure and arboreal niches • UNDERSTOREY → filtered light and humid microclimate	• SHRUB AND HERB LAYERS → near-ground resource partitioning • FOREST FLOOR → litter, detritivores and decomposer activity	• CAUTION → more layers do not prove universal stability • CANOPY → high light, wind exposure and arboreal niches

ECOSYSTEM STRUCTURE AND FUNCTION • TILE 2/4 • same master rows 2547-5781 of 10876 • continues below

06

STAGE 06

FOREST STRATIFICATION SECTION

FOREST STRATIFICATION SECTION

HIGH LIGHT, WIND EXPOSURE AND ARBOREAL NICHES

FILTERED LIGHT AND HUMID MICROCLIMATE

SHRUB AND HERB LAYERS

NEAR-GROUND RESOURCE PARTITIONING

FOREST FLOOR

LITTER, DETRITIVORES AND DECOMPOSER ACTIVITY

MORE LAYERS DO NOT PROVE UNIVERSAL STABILITY

CANOPY → high light, wind exposure and arboreal niches

UNDERSTOREY → filtered light and humid microclimate

SHRUB AND HERB LAYERS → near-ground resource partitioning

FOREST FLOOR → litter, detritivores and decomposer activity

SYSTEM / LEVEL / DEFINITION

• CANOPY → high light, wind exposure and arboreal niches

• UNDERSTOREY → filtered light and humid microclimate

STRUCTURE-FUNCTION MECHANISM

• SHRUB AND HERB LAYERS → near-ground resource partitioning

• FOREST FLOOR → litter, detritivores and decomposer activity

SCALE / EVIDENCE LIMIT

• CAUTION → more layers do not prove universal stability

• CANOPY → high light, wind exposure and arboreal niches

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CAUTION → more layers do not prove universal stability.

07

STAGE 07

PRODUCTIVITY ACCOUNTING

PRODUCTIVITY ACCOUNTING

GROSS PRIMARY PRODUCTIVITY

TOTAL PRODUCER FIXATION PER UNIT TIME

MINUS PRODUCER RESPIRATION

METABOLIC ENERGY USED BY PRODUCERS

NET PRIMARY PRODUCTIVITY

GROWTH AND PRODUCTION AVAILABLE ONWARD

STANDING CROP

GROSS PRIMARY PRODUCTIVITY → total producer fixation per unit time

MINUS PRODUCER RESPIRATION → metabolic energy used by producers

NET PRIMARY PRODUCTIVITY → growth and production available onward

STANDING CROP → living stock at a stated instant

RULE → a stock cannot substitute for a production rate

SYSTEM INPUT / STARTING CONDITION

• GROSS PRIMARY PRODUCTIVITY → total producer fixation per unit time

• MINUS PRODUCER RESPIRATION → metabolic energy used by producers

ECOLOGICAL PROCESS / TRANSFER

• NET PRIMARY PRODUCTIVITY → growth and production available onward

• STANDING CROP → living stock at a stated instant

OUTCOME / FEEDBACK / LIMIT

• RULE → a stock cannot substitute for a production rate

• GROSS PRIMARY PRODUCTIVITY → total producer fixation per unit time

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → a stock cannot substitute for a production rate.

08

STAGE 08

RESISTANCE AND RESILIENCE

RESISTANCE AND RESILIENCE

LIMITED DEPARTURE FROM PRIOR CONDITION DURING A DISTURBANCE

RECOVERY AFTER THE DISTURBANCE

MAY LOOK UNIFORM WHILE REMAINING SHOCK-SPECIFIC AND FRAGILE

DIVERSE WEB

MAY PROVIDE RESPONSE OPTIONS BUT NO UNIVERSAL GUARANTEE

NAME THE SHOCK, RESPONSE VARIABLE AND TIME FRAME

ECOLOGICAL UNIT / PARAMETER

RESISTANCE → limited departure from prior condition during a disturbance

RESILIENCE → recovery after the disturbance

DECISIVE DISTINCTION

MONOCULTURE → may look uniform while remaining shock-specific and fragile

DIVERSE WEB → may provide response options but no universal guarantee

SCALE / STATUS / EXCEPTION

ANSWER → name the shock, response variable and time frame

RESISTANCE → limited departure from prior condition during a disturbance

ECOLOGICAL UNIT / PARAMETER

• RESISTANCE → limited departure from prior condition during a disturbance

• RESILIENCE → recovery after the disturbance

DECISIVE DISTINCTION

• MONOCULTURE → may look uniform while remaining shock-specific and fragile

• DIVERSE WEB → may provide response options but no universal guarantee

SCALE / STATUS / EXCEPTION

• ANSWER → name the shock, response variable and time frame

• RESISTANCE → limited departure from prior condition during a disturbance

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ANSWER → name the shock, response variable and time frame.

09

STAGE 09

SERVICES AND CONTRIBUTIONS

SERVICES AND CONTRIBUTIONS

MATERIAL OUTPUTS SUCH AS FOOD, FIBRE AND WATER

FLOOD BUFFERING, POLLINATION AND CLIMATE MODERATION

SOIL FORMATION, PRIMARY PRODUCTION AND NUTRIENT CYCLING

SPIRITUAL, AESTHETIC, RECREATIONAL AND KNOWLEDGE VALUES

INCLUDES RELATIONAL AND INDIGENOUS OR LOCAL-KNOWLEDGE VALUES

ECOLOGICAL UNIT / PARAMETER

PROVISIONING → material outputs such as food, fibre and water

REGULATING → flood buffering, pollination and climate moderation

DECISIVE DISTINCTION

SUPPORTING → soil formation, primary production and nutrient cycling

CULTURAL → spiritual, aesthetic, recreational and knowledge values

SCALE / STATUS / EXCEPTION

NCP → includes relational and indigenous or local-knowledge values

PROVISIONING → material outputs such as food, fibre and water

ECOLOGICAL UNIT / PARAMETER

• PROVISIONING → material outputs such as food, fibre and water

• REGULATING → flood buffering, pollination and climate moderation

DECISIVE DISTINCTION

• SUPPORTING → soil formation, primary production and nutrient cycling

• CULTURAL → spiritual, aesthetic, recreational and knowledge values

SCALE / STATUS / EXCEPTION

• NCP → includes relational and indigenous or local-knowledge values

• PROVISIONING → material outputs such as food, fibre and water

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NCP → includes relational and indigenous or local-knowledge values.

09

STAGE 09 SERVICES AND CONTRIBUTIONS

SERVICES AND CONTRIBUTIONS

MATERIAL OUTPUTS SUCH AS FOOD, FIBRE AND WATER

FLOOD BUFFERING, POLLINATION AND CLIMATE MODERATION

SOIL FORMATION, PRIMARY PRODUCTION AND NUTRIENT CYCLING

SPIRITUAL, AESTHETIC, RECREATIONAL AND KNOWLEDGE VALUES

INCLUDES RELATIONAL AND INDIGENOUS OR LOCAL-KNOWLEDGE VALUES

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
PROVISIONING → material outputs such as food, fibre and water	SUPPORTING → soil formation, primary production and nutrient cycling	NCP → includes relational and indigenous or local-knowledge values
REGULATING → flood buffering, pollination and climate moderation	CULTURAL → spiritual, aesthetic, recreational and knowledge values	PROVISIONING → material outputs such as food, fibre and water
ECOLOGICAL UNIT / PARAMETER • PROVISIONING → material outputs such as food, fibre and water • REGULATING → flood buffering, pollination and climate moderation	DECISIVE DISTINCTION • SUPPORTING → soil formation, primary production and nutrient cycling • CULTURAL → spiritual, aesthetic, recreational and knowledge values	SCALE / STATUS / EXCEPTION • NCP → includes relational and indigenous or local-knowledge values • PROVISIONING → material outputs such as food, fibre and water

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NCP → includes relational and indigenous or local-knowledge values.

10

STAGE 10 CARRYING-CAPACITY ANSWER SPINE

CARRYING-CAPACITY ANSWER SPINE

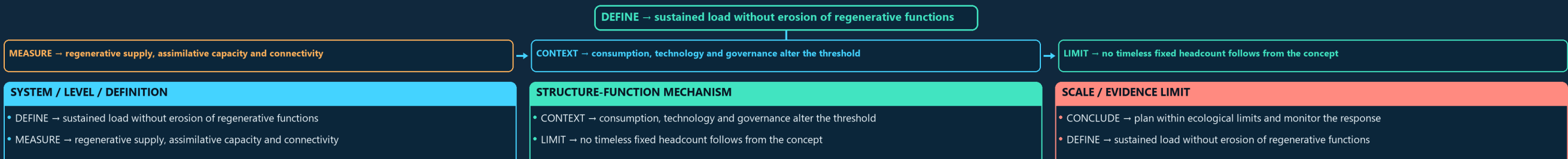
SUSTAINED LOAD WITHOUT EROSION OF REGENERATIVE FUNCTIONS

REGENERATIVE SUPPLY, ASSIMILATIVE CAPACITY AND CONNECTIVITY

CONSUMPTION, TECHNOLOGY AND GOVERNANCE ALTER THE THRESHOLD

NO TIMELESS FIXED HEADCOUNT FOLLOWS FROM THE CONCEPT

PLAN WITHIN ECOLOGICAL LIMITS AND MONITOR THE RESPONSE



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONCLUDE → plan within ecological limits and monitor the response.

11

STAGE 11 EVIDENCE AND INSTITUTION FIREWALL

EVIDENCE AND INSTITUTION FIREWALL

BASIC OWNER

DEFINITIONS, MECHANISMS AND VERIFIED PYQ CONCEPTS

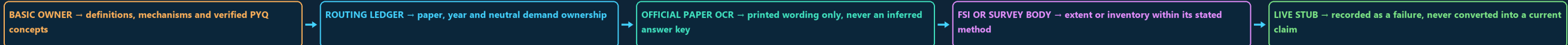
ROUTING LEDGER

PAPER, YEAR AND NEUTRAL DEMAND OWNERSHIP

OFFICIAL PAPER OCR

PRINTED WORDING ONLY, NEVER AN INFERRED ANSWER KEY

FSI OR SURVEY BODY



ECOLOGICAL UNIT / PARAMETER • BASIC OWNER → definitions, mechanisms and verified PYQ concepts • ROUTING LEDGER → paper, year and neutral demand ownership	DECISIVE DISTINCTION • OFFICIAL PAPER OCR → printed wording only, never an inferred answer key • FSI OR SURVEY BODY → extent or inventory within its stated method	SCALE / STATUS / EXCEPTION • LIVE STUB → recorded as a failure, never converted into a current claim • BASIC OWNER → definitions, mechanisms and verified PYQ concepts
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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIVE STUB → recorded as a failure, never converted into a current claim.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

MOEFCC OPERATIONALISES ECOSYSTEM CONCEPTS THROUGH SPECIES/HABITAT LAW (TOPICS 06-10)

WII, BSI, ZSI GENERATE THE BASELINE ECOLOGICAL DATA (SPECIES

IMPLEMENTATION GAP

ECOSYSTEM-SERVICES VALUE IS RARELY MONETISED IN INDIAN PROJECT

FOREST/TREE COVER STATISTICS (ISFR) MEASURE CANOPY DENSITY THRESHOLDS, NOT

"ECOSYSTEM HEALTH" HAS NO SINGLE AGREED METRIC; EXAMINERS REWARD

KEystone-SPECIES ARGUMENTS (E.G., APEX PREDATOR REMOVAL TRIGGERING TROPHIC CASCADE)

OPTIONAL ECOLOGICAL PROCESS DEPTH

- MoEFCC operationalises ecosystem concepts through species/habitat law (Topics 06-10),
- WII, BSI, ZSI generate the baseline ecological data (species inventories, habitat
- CAUTION: Implementation gap: ecosystem-services value is rarely monetised in Indian project
- CAUTION: Forest/tree cover statistics (ISFR) measure canopy density thresholds, not species

SCALE, PARAMETER OR STATUS LIMIT

- CAUTION: "Ecosystem health" has no single agreed metric; examiners reward candidates who name
- CAUTION: Keystone-species arguments (e.g., apex predator removal triggering trophic cascade) are
- Only about 10% of energy is transferred on average from one trophic level to the next
- Net Primary Productivity (NPP) = Gross Primary Productivity (GPP) – respiratory loss by

COMPARATIVE RESTORATION NUANCE

- The Millennium Ecosystem Assessment's four-fold service classification (provisioning,
- Ecotones show the edge effect — higher species density and variety than either bordering
- IPBES reframed the MEA's "ecosystem services" as Nature's Contributions to People
- A niche is not a single attribute: it decomposes into habitat, trophic (food) and

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.